Convex optimization for machine learning – TD2

Olivier Fercoq
Bruno Costacèque

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1 Exercises

Exercise 1. (Proximal operator) Let $f : \mathbb{R}^n \to \mathbb{R} \cup +\infty$ be a convex lower-semicontinuous function such that $\mathrm{dom} f \neq \emptyset$.

1. Recall the definition of the domain of a convex function.

2. It is possible to prove (but we do not ask you to do it) that $\exists \mathbf{x}_0 \in \mathrm{dom} f$ such that $\partial f(\mathbf{x}_0) \neq \emptyset$. Using this information, show that there exists $\alpha \in \mathbb{R}$ and $\mathbf{w} \in \mathbb{R}^n$ such that for all $\mathbf{x}$ we have $f(\mathbf{x}) \geq \alpha + \langle \mathbf{w}, \mathbf{x} \rangle$.

3. Let us fix $\mathbf{x} \in \mathbb{R}^n$. Let us define $g : \mathbf{y} \mapsto f(\mathbf{y}) + \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2$. Show that g is strongly convex.

4. Show that $\lim\limits_{\|\mathbf{y}\| \to \infty} g(\mathbf{y}) = +\infty$.

5. Show that g has a minimizer and that it is unique. We will denote this minimizer as $\mathrm{prox}_f(\mathbf{x})$. The function $\mathrm{prox}_f : \mathbb{R}^n \to \mathbb{R}^n$ is called the *proximal operator* of f.

Exercise 2. Let us denote:

$$\mathrm{prox}_g(\mathbf{y}) = \arg\min_{\mathbf{x} \in \mathcal{X}} \Big\{ g(\mathbf{x}) + \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2 \Big\}$$

the proximal operator of g at $\mathbf{y}$. Fix $\gamma > 0$. Show that if f and g are convex, and f differentiable, then the fixed points of the nonlinear equation

$$\mathbf{x} = \mathrm{prox}_{\gamma g}\big(\mathbf{x} - \gamma \nabla f(\mathbf{x})\big) \tag{1}$$

are the minimizers of the function $F = f + g$.

Exercise 3. (Taylor-Lagrange inequality) The goal of this exercise is to prove Taylor-Lagrange inequality. This is a fundamental inequality for the study of gradient descent and related methods.

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function whose gradient is L-Lipschitz *i.e.* $\|\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})\| \leq L\|\mathbf{y} - \mathbf{x}\|$ for all $\mathbf{x}, \mathbf{y}$.

1. Prove that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\big\langle \nabla f(\mathbf{y}) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \big\rangle \leq L\|\mathbf{y} - \mathbf{x}\|^2$.

2. Set $\varphi(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ for all $t \in [0, 1]$. Prove that:

$$f(\mathbf{y}) - f(\mathbf{x}) - \big\langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \big\rangle = \varphi(1) - \varphi(0) - \varphi'(0).$$

3. Deduce that:

$$f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle = \int_0^1 \langle \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \mathrm{d}t$$

4. Using the first question, conclude that

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$